

Optimization Problems.

Here we will study applications of the first and second derivative test that is, using maximum and minimum.

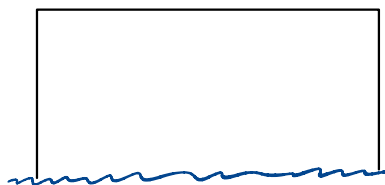
The methods we have learned in max/min section for finding extreme values which have many applications in the real life.

Steps in Solving Optimization Problems:

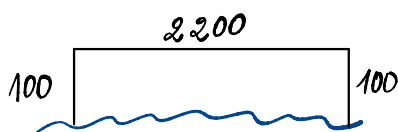
1. Understand the problem: The first step is to read the problem carefully until it's clearly understood.
2. Draw a Diagram.
3. Introduce Notation (Using new variables).
4. Use these variables and obtain a relationship between them.
5. Obtain a function in only one variable.

Example 1. A farmer has 2400 ft of fencing and wants to fence off a rectangular field that borders a straight river. He needs no fence along the river. What are the dimensions of the field that has the largest area?

Sln: step 2.

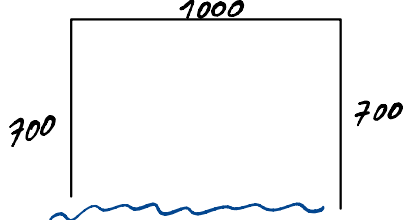


Notice that: 



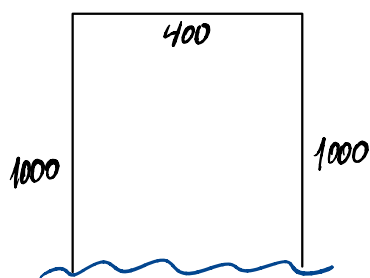
$$A = (100)(2200) = 220,000 \text{ ft}^2$$

$$P = 100 + 2200 + 100 = 2400 \text{ ft} \checkmark$$



$$A = (700)(1000) = 700,000 \text{ ft}^2$$

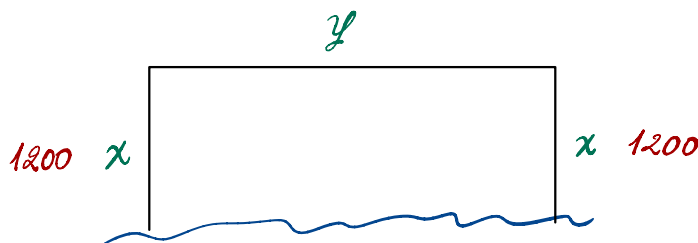
$$P = 700 + 1000 + 700 = 2400 \text{ ft} \checkmark$$



$$A = (1000)(400) = 400,000 \text{ ft}^2$$

$$P = 1000 + 400 + 1000 = 2400 \checkmark$$

Step 3.



Restriction:

$$x + y + x = 2400$$

We are looking for the largest area:

my "function" is the area function:

$$A = x \cdot y$$

We have two variables.

Using my restriction:

$$x + y + x = 2400$$

$$2x + y = 2400$$

$$y = 2400 - 2x$$

then

$$A(x) = x \cdot (2400 - 2x)$$

$$A(x) = 2400x - 2x^2, \quad 0 \leq x \leq 1200$$

Thm from 2.24: If f is a continuous function on a closed interval $I = [a, b]$ then f has a abs. max and abs. minimum.

Prmk. If f is defined on $(I=[a,b])$ a closed interval, then the critical points are points s.t $f'(x_0)=0$ and $f'(x_0)=DNE$ and $x_0=a, x_0=b$.

Solving the problem: $A(x) = 2400x - 2x^2; 0 \leq x \leq 1200$.

$$A'(x) = 2400 - 4x. \text{ So, } 2400 - 4x = 0 \Leftrightarrow x = \frac{2400}{4} = 600.$$

is 600 the only critical point? *No*, b/c the function $A(x)$ is defined on a closed interval.

The critical points: $x=0, x=600, x=1200$

$$\text{at } x=0 \Rightarrow A(0) = 0(2400 - 2(0)) = 0.$$

$$\begin{aligned} \text{at } x=600 \Rightarrow A(600) &= 600(2400 - 2(600)) = 600(2400 - 1200) = 600 \cdot 1200 \\ &= 720,000 \text{ ft}^2 \end{aligned}$$

$$\text{at } x=1200 \Rightarrow A(1200) = 1200(2400 - 2(1200)) = 0.$$

So, the maximum area is $720,000 \text{ ft}^2$ when $x=600 \text{ ft}$.

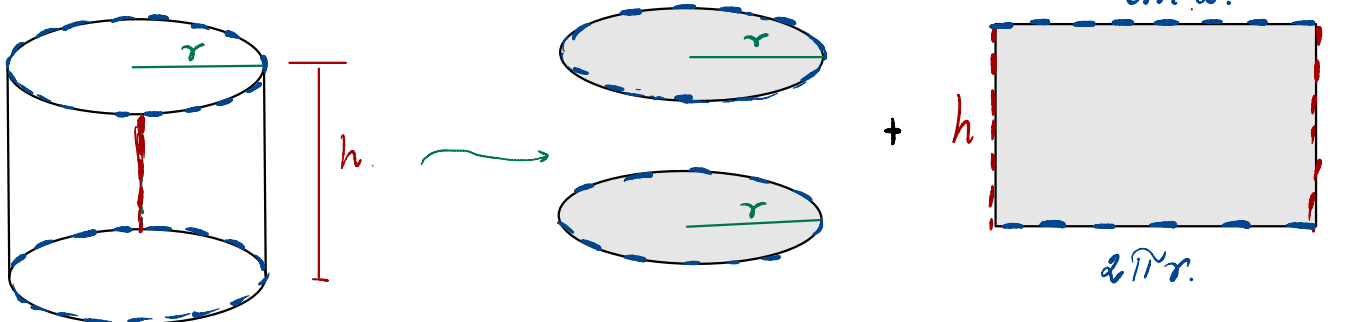
We need to find "y"

$$y = 2400 - 2(600) = 1200 \text{ ft}.$$

So, the rectangular field should be 600 ft deep and 1200 ft wide.

Example 2. A cylindrical can is to be made to hold 1L of oil. Find the dimensions that will minimize the cost of the metal to manufacture the can.

Sln.



then we are looking for the minimum of the area:

$$A = 2\pi r^2 + (2\pi r)h = 2\pi r^2 + 2\pi r h.$$

we have restriction that the volume is $1L = 1000 \text{ cm}^3$.

$$V = \pi r^2 \cdot h = 1000.$$

then solving for h:

$$h = \frac{1000}{\pi r^2}$$

$$A = 2\pi r^2 + 2\pi r \left(\frac{1000}{\pi r^2} \right)$$

$$A(r) = 2\pi r^2 + \frac{2000}{r}$$

then solving for r:

$$r = \sqrt{\frac{1000}{h}}$$

$$A = 2\pi \left(\sqrt{\frac{1000}{h}} \right)^2 + 2\pi \sqrt{\frac{1000}{h}} \cdot h$$

$$A(h) = \frac{2000\pi}{h} + 2\pi \sqrt{1000} \cdot \sqrt{h}.$$

Solving:

$$A'(r) = 4\pi r - \frac{2000}{r^2} = \frac{4\pi r^3 - 2000}{r^2}.$$

the only critical point is

$$r = \sqrt[3]{\frac{2000}{4\pi}} = \sqrt[3]{\frac{500}{\pi}}.$$

$$A''(r) = 4\pi + \frac{4000}{r^3} \quad ; \quad A''\left(\sqrt[3]{\frac{500}{\pi}}\right) = 4\pi + \frac{4000}{\left(\sqrt[3]{\frac{500}{\pi}}\right)^3} = 4\pi + \frac{4000\pi}{500} > 0$$

So, by the second derivative test we have a minimum at $r = \sqrt[3]{\frac{500}{\pi}}.$

$$\text{and } h = \frac{1000}{\pi \left(\frac{500}{\pi}\right)^{2/3}}.$$